Pulmonary Nodule False Positive Reduction from CT Images Using 3D CNN on the LUNA16 Dataset

Lab Report

***Abstract*—We apply a 3D CNN to the false positive reduction task on the LUNA16 dataset, classifying 32^3-voxel patches centered on candidate locations as nodule or non-nodule. Using Focal Loss and balanced sampling to handle extreme class imbalance ($\sim$1:484), the model achieves an AUC-ROC of 0.9978, sensitivity of 0.9941, and specificity of 0.9616.**

I. Introduction

Pulmonary nodules are small lung masses that may indicate early-stage lung cancer. Automated CAD systems can improve CT screening efficiency but typically produce many false positives. The LUNA16 challenge [1] addresses this through a false positive reduction track: classifying pre-computed candidate locations as true nodules or false positives. We tackle this task using a 3D CNN and evaluate with standard classification metrics (AUC-ROC, sensitivity, specificity, F1).

II. Exploratory Data Analysis

The LUNA16 dataset contains 888 CT scans with annotations from four radiologists. We use subsets 0–4 (445 scans) available on Kaggle. Table I summarizes key statistics; the `candidates_V2.csv` file provides 754,975 candidate locations with a $\sim$1:484 class imbalance (Fig. 1). Nodule diameters are right-skewed (median 6.4 mm, mean 8.3 mm). CT scans have typical axial resolution of 512×512 with 0.7 mm in-plane spacing and 0.625–2.5 mm slice thickness.

TABLE I
LUNA16 Dataset Statistics

Metric	Value
CT scans (available subsets 0–4)	445
Annotated nodules (available)	615
Candidates V2 (total)	754,975
Positive / Negative candidates	1,557 / 753,418
Class imbalance ratio	$\sim$1:484
Nodule diameter mean / median	8.3 / 6.4 mm

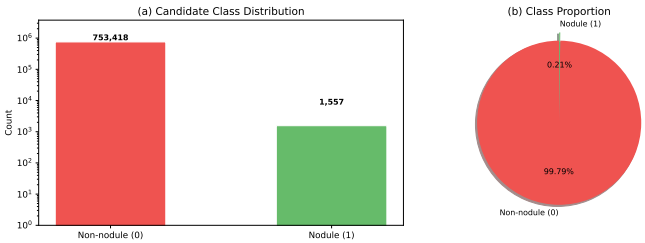

Fig. 1. Candidate class distribution showing extreme imbalance ($\sim$1:484). (a) Log-scale bar chart. (b) Proportion.

Fig. 2 illustrates the visual similarity between nodule and non-nodule patches that makes classification challenging.

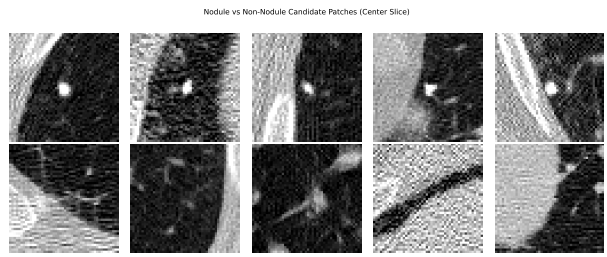

Fig. 2. Nodule (top) vs non-nodule (bottom) candidate patches. Center axial slices shown.

III. Methods

A. Data Pipeline and Augmentation

For each candidate, we load the CT scan via SimpleITK, convert world coordinates to voxel indices, extract a 32^3 patch (zero-padded at boundaries), and normalize HU values from $[-1000, 400]$ to $[0, 1]$. To address memory constraints, negative candidates are subsampled to a 1:20 ratio, reducing $\sim$377K to $\sim$16K total patches. Subsets 0–3 are used for training and subset 4 for validation. A `WeightedRandomSampler` ensures balanced 1:1 training epochs. Augmentation includes random 3D flips (50%), 90° axial rotations, and Gaussian noise ($\sigma = 0.01$, 50%).

B. 3D CNN Architecture

The model ($\sim$7.0M parameters) has four convolutional blocks followed by a three-layer classifier:

- **Blocks 1–3**: Conv3d $\times$2 + BN + ReLU + MaxPool(2^3), channels $1 \to 64 \to 128 \to 256$
- **Block 4**: Conv3d($256 \to 512$) + BN + ReLU + GlobalAvgPool
- **Classifier**: FC($512 \to 128$) + Drop(0.5) $\to$ FC($128 \to 64$) + Drop(0.3) $\to$ FC($64 \to 1$)

C. Training and Evaluation

We use Focal Loss (α=0.75, γ=2.0) with Adam (lr $= 10^{-3}$, wd $= 10^{-4}$), ReduceLROnPlateau scheduler, mixed-precision training on an NVIDIA P100, batch size 32, and early stopping (patience 8 on val AUC). Evaluation metrics include AUC-ROC, and sensitivity/specificity/F1 at the optimal Youden's J threshold ($J = \text{TPR} - \text{FPR}$).

IV. Results

Table II summarizes the validation performance. The model achieves AUC-ROC of 0.9978, indicating near-perfect candidate-level discrimination. At the optimal threshold, sensitivity is 99.41% with 96.16% specificity. The confusion matrix (Fig. 3) shows most errors are false positives.

TABLE II
Validation Results (Subset 4)

Metric	Value
AUC-ROC	**0.9978**
Sensitivity	0.9941
Specificity	0.9616
F1-Score	0.7226

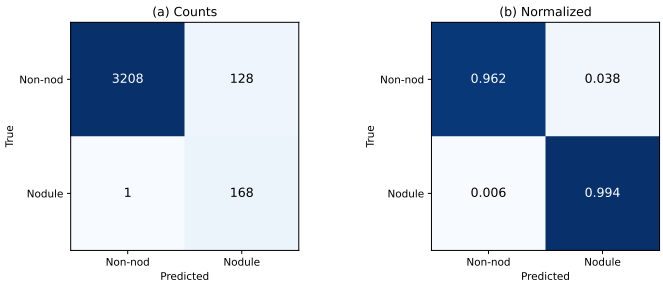

Fig. 3. Confusion matrix: (a) raw counts, (b) row-normalized.

A. Qualitative Results

Fig. 4 shows sample predictions. The model correctly identifies most nodules with high confidence while assigning low probabilities to non-nodule candidates.

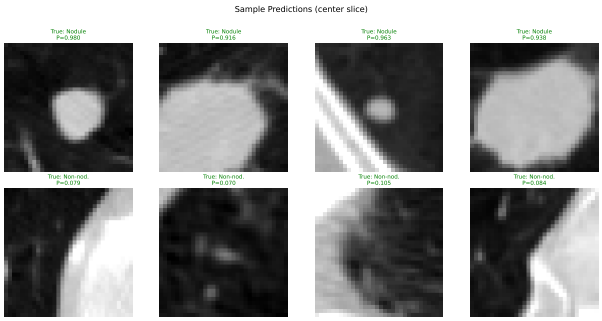

Fig. 4. Sample predictions. Top: true nodules; bottom: true non-nodules. Green = correct, red = incorrect.

V. Discussion

Class imbalance handling. Focal Loss (γ=2.0) combined with balanced sampling effectively addresses the $\sim$1:484 imbalance by down-weighting easy negatives.

Limitations. Only subsets 0–4 are used (half the data), and memory-constrained subsampling limits hard negative diversity. Future work could explore 3D ResNets, attention mechanisms, larger patches, and evaluation on the full dataset.

VI. Conclusion

A 3D CNN with Focal Loss and balanced sampling achieves AUC-ROC of 0.9978 with 99.41% sensitivity and 96.16% specificity on the LUNA16 false positive reduction task. Training on the full dataset with deeper architectures could further improve performance.

References

[1] A. A. A. Setio *et al.*, "Validation, comparison, and combination of algorithms for automatic detection of pulmonary nodules in computed tomography images: The LUNA16 challenge," *Medical Image Analysis*, vol. 42, pp. 1–13, 2017.